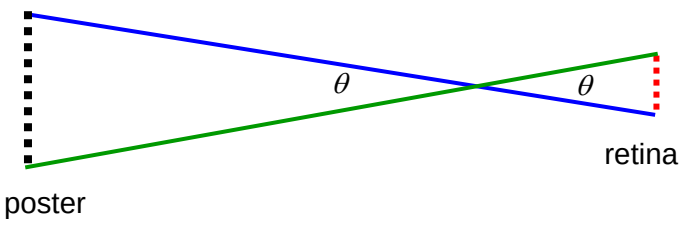


Q6	
(a)(i)	A stationary wave is formed when <u>two progressive waves of the same type</u> of <u>equal amplitude</u> , equal <u>frequency</u> (or wavelength) and <u>same speed</u> , meet when they travel in <u>opposite directions</u> .
(a)(ii)1	For fundamental frequency, length of string $L = \frac{\lambda}{2}$ Consider length of string L, and speed of each progressive wave, $v = f \lambda$ $405 = (622) 2L$ $L = 0.326 \text{ m (3 s.f.)}$
(a)(ii)2	$v = \omega \sqrt{x_0^2 - x^2}$ $v_{\max} = \omega x_0$ $= 2\pi(622) 3.3 \times 10^{-3}$ $= 12.9 \text{ m s}^{-1} \text{ (3 s.f.)}$ $= 12.9 \text{ m s}^{-1} \text{ (3 s.f.)}$
(a)(iii)1	$I = \frac{P}{2\pi r^2}$ $= \frac{200}{2\pi(10.0)^2}$ $= 0.318 \text{ W m}^{-2} \text{ (3 s.f.)}$
(a)(ii)	Possible reasons (any 2)
	• If the 2 identical loudspeakers are far apart, the path difference between the 2 waves is significant and their amplitudes are significantly different at all points (except the mid point/middle section). • Waves reflections from surfaces will superpose with the 2 waves from the loudspeakers. • There is an obstacle between the 2 loudspeakers, and the 2 waves are unable to meet. • Air in the medium is not stationary (e.g. if wind is blowing from one speaker to another, one wave has a smaller, and one has a larger frequency than the original frequency).
(b)(i)	For single slit diffraction, at the 1st minima $m = 1$, $\sin \theta = \frac{m \lambda}{b} \quad \dots\dots\dots (1)$

	For small angle approximation, $\sin \theta \approx \tan \theta \dots\dots\dots (2)$ $\tan \theta = \sin \theta$ $\tan \theta = \frac{\lambda}{b}$ $\frac{w/2}{L} = \frac{\lambda}{b}$ $\frac{4.80 \times 10^{-6} / 2}{17.0 \times 10^{-3}} = \frac{\lambda}{3.0 \times 10^{-3}}$ $\lambda = 4.24 \times 10^{-7} \text{ m}$ $= 424 \text{ nm (3 s.f.)}$ OR Pythagoras Theorem can be used to derive equation for θ.
(b)(ii)1	Two images are <u>just resolved</u> when the <u>central maximum of one image falls on the first minimum of the other image.</u>
(b)(ii)2	 Applying Rayleigh Criterion, for <u>minimum</u> wavelength $\theta_R = \frac{\lambda_{\min}}{b} \dots\dots\dots (1)$ $= \frac{380 \times 10^{-9}}{3.00 \times 10^{-3}}$ $= 1.27 \times 10^{-4} \text{ rad (3 s.f.)}$
(b)(ii)3	By geometry and small angle approximation, for minimum distance between 2 adjacent pixels, $\theta_{\min} = \frac{d_{\min}}{D} \dots\dots\dots (2)$ $d_{\min} = \frac{72 \times 10^{-2} \text{ m}}{710}$ $\theta_{\min} = \frac{d_{\min}}{D}$ $= \left(\frac{72 \times 10^{-2}}{710} \right)$ $= 1.5$ $= 6.761 \times 10^{-4} \text{ rad}$ Since $\theta_{\min} > 2.91 \times 10^{-4} \text{ rad}$, the observer can distinguish between any two adjacent blue pixels of the smaller distance and smallest wavelength of blue.

(b)(ii)4	Slit width increases, angular resolution decreases, hence minimum distance increases.
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